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1. Which of the following is not a way for a derivative to fail to exist?

- a. cusp c. vertical tangent
 b. horizontal tangent d. discontinuity

2. An object moves in a straight line with its position at time t seconds given by $s(t) = -3t + t^2$, where s is measured in metres. Determine the velocity of the object when $t = 3$.

- a. -3 m/s c. 3 m/s
 b. 0 m/s d. 6 m/s

3. All but one of the functions is differentiable for all real values of x . Which function is not differentiable for at least one real value of x ?

- a. $f(x) = x^2 + 1$ c. $h(x) = |x|$
 b. $g(x) = \frac{1}{x^2 + 1}$ d. $j(x) = x^3 - 3x$

4. Determine the value of k for which $f(x) = 4x^2 - kx + 6$ has a horizontal tangent at $x = \frac{1}{2}$.

- a. 1 c. 4
 b. 2 d. 8

$$0 = 8\left(\frac{1}{2}\right) - k$$

$$k = 4$$

5. Find $h'(2)$ given $h(x) = f(g(x))$, $f(u) = 2u^3 - 2u$, $g(2) = 3$, and $g'(2) = 4$

$$h(x) = f(g(x))$$

$$f(u) = 2u^3 - 2u$$

$$g(2) = 3$$

$$g'(2) = 4$$

$$h'(x) = f'(g(x))g'(x)$$

$$f'(u) = 6u^2 - 2$$

$$h'(x) = (6g(x)^2 - 2)g'(x)$$

$$h'(2) = (6(g(2))^2 - 2)g'(2)$$

$$h'(2) = (6(3)^2 - 2)(4)$$

$$= (52)(4)$$

$$h'(2) = 208$$

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6. Find the point(s) on the graph of $f(x) = \frac{x}{x^2+1}$ where the tangent is horizontal.

$$f(x) = x(x^2+1)^{-1}$$

$$f'(x) = 1(x^2+1)^{-1} + x(-1)(x^2+1)^{-2}(2x)$$

$$= \frac{1}{x^2+1} - \frac{2x^2}{(x^2+1)^2}$$

$$= \frac{(x^2+1) - 2x^2}{(x^2+1)^2}$$

$$= \frac{-x^2+1}{(x^2+1)^2}$$

$$f'(x) = 0$$

$$0 = -x^2 + 1$$

$$x^2 = 1$$

$$x = \pm 1$$

$$P(1, \frac{1}{2})$$

$$P(-1, -\frac{1}{2})$$

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7. If $y = u^3 + u^2 - 1$, where $u = \frac{1}{1-x}$, determine $\frac{dy}{dx}$ at $x = 2$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{du} = 3u^2 + 2u$$

$$\frac{du}{dx} = \frac{1}{(1-x)^2}$$

When $x = 2$,

$$u = \frac{1}{1-2} = -1$$

$$u = -1$$

$$= \frac{1}{(1-x)^2}$$

$$\frac{dy}{dx} = (3u^2 + 2u) \left(\frac{1}{(1-x)^2} \right)$$

$$\frac{dy}{dx} \bigg|_{x=2} = (3(-1)^2 + 2(-1)) \left(\frac{1}{(1-2)^2} \right)$$

$$= (1) \left(\frac{1}{(-1)^2} \right)$$

$$= 1$$

$$u = \frac{1}{1-x}$$

$$u' = -\frac{1}{(1-x)^2}(-1)$$

$$= \frac{1}{(1-x)^2}$$

$$y' = (3(\frac{1}{1-x})^2 + 2(\frac{1}{1-x})) \left(\frac{1}{(1-x)^2} \right)$$

$$= (3(\frac{1}{1-x})^2 + 2(\frac{1}{1-x})) \left(\frac{1}{(1-x)^2} \right)$$

$$= (3+2)(1)$$

$$= 1$$

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8. Determine the value of b if the slope of the tangent to $f(x) = (3x^2 + 1)(2x^2 + b)$ at $x = -1$ is -16 .

$$f'(-1) = -16$$

$$f(x) = (3x^2 + 1)(2x^2 + b)$$

$$f'(x) = (6x)(2x^2 + b) + (3x^2 + 1)(4x)$$

$$= 12x^3 + 6xb + 12x^3 + 4x$$

$$= 24x^3 + 4x + 6xb$$

$$-16 = 24(-1)^3 + 4(-1) + 6(-1)b$$

$$-16 = -24 - 4 - 6b$$

$$12 = -6b$$

$$b = -2$$

Check

$$f(x) = (3x^2 + 1)(2x^2 + 2)$$

$$= 6x^4 + 4x^2 + 2x^2 + 2$$

$$f'(x) = 24x^3 + 8x$$

$$f'(-1) = -24 + 8$$

$$= -16$$

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9. Determine the values of a , b , and c for $f(x) = ax^2 + bx + c$ so that $f'(x) = 6x - 3$ and $f(2) = -1$.

$$S(x) = ax^2 + bx + c$$

$$S'(x) = 6x - 3$$

$$-1 = a(2)^2 + b(2) + c$$

$$-1 = 4a + 2b + c$$

$$c = -1 - 4a - 2b$$

$$S'(x) = 2ax + b$$

$$6x - 3 = 2ax + b$$

$$b = -3$$

$$2a = 6$$

$$a = 3$$

$$c = -1 - 4(3) - 2(-3)$$

$$= -1 - 12 + 6$$

$$c = -7$$

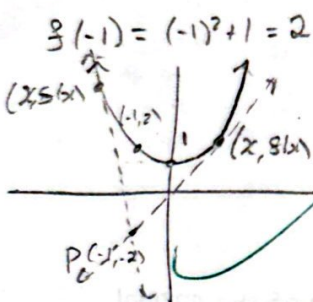
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Check
 $S(2) = 3(4) - 3(2) - 7$
 $= 12 - 6 - 7$
 $= -1$

$$S'(x) = 6x - 3$$

$$S(x) = 3x^2 - 3x - 7$$

10. Find the equation(s) of the tangent line(s) to the curve $f(x) = x^2 + 1$ passing through the point $P(-1, -2)$. Draw a rough diagram to illustrate the situation.



$$S(-1) = (-1)^2 + 1 = 2$$

$$S(x) = x^2 + 1$$

$$S'(x) = 2x = m$$

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{S(x_1) - (-2)}{x_1 - (-1)}$$

$$= \frac{(x_1^2 + 1) + 2}{x_1 + 1}$$

$$= \frac{x_1^2 + 3}{x_1 + 1}$$

$$2x = \frac{x^2 + 3}{x + 1}$$

$$2x(x + 1) = x^2 + 3$$

$$2x^2 + 2x = x^2 + 3$$

$$x^2 + 2x - 3 = 0$$

$$(x + 3)(x - 1) = 0$$

$$x = -3, 1$$

$$S'(-3) = 2(-3) = -6$$

$$-2 = -6(-1) + b$$

$$b = -8$$

$$y = -6x - 8$$

$$P(-1, -2)$$

$$S'(1) = 2(1) = 2$$

$$-2 = 2(-1) + b$$

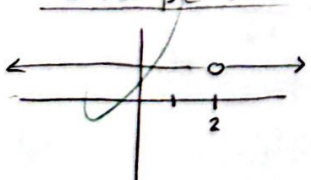
$$b = 0$$

$$y = 2x$$

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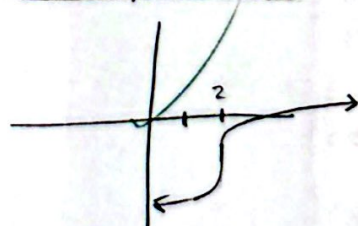
11. Illustrate and briefly describe two situations where a function does not have a derivative at $x=2$.

Example 1:



Graph is discontinuous at $x=2$, (point discrep). $S(x)$ DNE at $x=2$ so therefore there is no slope and it is non-differentiable.

Example 2:



Graph has a vertical tangent at $x=2$.
 $\lim_{x \rightarrow 2^-} S'(x) = \infty$ $\lim_{x \rightarrow 2^+} S'(x) = \infty$
 $\therefore \lim_{x \rightarrow 2} S'(x) = \text{DNE}$

And since both lim from left and right equal $+\infty$, graph has a vertical tangent.

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12. Indicate whether the following are True or False.

a) the slope of the tangent to $f(x) = \sqrt{1-2x}$ at $(-4, 3)$ is $-\frac{2}{3}$

$$f'(x) = \frac{1}{2}(1-2x)^{-1/2}(-2) = \frac{-1}{\sqrt{1-2x}} \quad f'(-4) = \frac{-1}{\sqrt{9}} = -\frac{1}{3}$$

b) you can use product rule to differentiate a rational function

$$h(x) = \frac{f(x)}{g(x)} = f(x)g(x)^{-1}$$

c) if f and g are differentiable, then the derivative of $f(x) + g(x)$ is $f'(x) + g'(x)$.

d) If a function is differentiable at point $(a, f(a))$, then $f(x)$ is continuous at $x = a$.

13. Given $y = \frac{x^{3a+3b}}{x^{a-2b}}$, and a, b are constants:

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

a) Determine $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{(3a+3b)x^{3a+3b-1}(x^{a-2b}) - (x^{3a+3b})(a-2b)x^{a-2b-1}}{(x^{a-2b})^2}$$

$$= \frac{(3a+3b)(x^{3a+3b})(x^{-1})(x^{a-2b}) - (x^{3a+3b})(a-2b)(x^{a-2b})(x^{-1})}{(x^{a-2b})^2}$$

$$= \frac{(x^{3a+3b})(x^{-1}) \cancel{(x^{a-2b})} [(3a+3b) - (a-2b)]}{(x^{a-2b})^2}$$

$$= \frac{x^{3a}x^{3b}x^{-1}(3a+3b-a+2b)}{x^a x^{-2b}}$$

$$= x^{2a}x^{5b}x^{-1}(2a+5b)$$

$$\frac{dy}{dx} = x^{2a+5b-1}(2a+5b)$$

b) Determine the slope of the normal

When $x=2$, $a=-1$ and $b=1$.

$$\left. \frac{dy}{dx} \right|_{x=2, a=-1, b=1} = (2)^{2(-1)+5(1)-1} (2(-1)+5(1))$$

$$= 2^2(3)$$

$$= 4(3)$$

$$m = 12$$

$$\text{Normal} = -\frac{1}{m}$$

$$m_{\text{normal}} = -\frac{1}{12}$$